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MILITARY SPECIFICATION

ALTIMETER, PRESSURIZED COMPARTMENT INTEGRALLY LIGHTED, 0 - 50,000 FEET, 1-7/8 INCH DIAL

* This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a single pointer, integrally lighted, 0-to 50,000-foot range cabin altimeter.

* 1.2 Classification. Altimeters shall be of the following types, as specified (see 6.2);

* AAU-3A Cover glass coated with an antireflection coating per MIL-C-675.

* AAU-3A/A Cover glass coated with multilayer antireflection coating of high efficiency.

2. APPLICABLE DOCUMENTS.

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-675	Coating of Glass Optical Elements (Anti-Reflection)
* DOD-D-1000	Drawing, Engineering and Associated List

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the selection Of
MIL-STD-794	Parts and Equipment Procedures for Packaging and Packing of.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS-28105	Cover, Glass - Aircraft Instrument Dial
MS-33558	Numeral and Letter, Aircraft Instrument Dial, Standard Form Of
MS-33585	Pointer, Dial, Standard Design of Aircraft Instrument
MS-33638	Case, Instrument, Flange-Mounted, Aircraft
MS-33649	Boss, Fluid Connection Internal straight Thread

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B-633-78 - Zinc on Iron and steel, electro-deposited coatings of.

ASTM D 3951 Standard Practice for Commerical Packaging

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* (Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The altimeter furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable products list.

3.2 Selection of specification and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials and parts.

3.3.1 Fungus-resistant materials. Materials that are not nutrients for fungi shall be used to the greatest extent practicable. Where fungi-nutrient materials must be used, they shall be treated with a fungicidal agent as approved by the procuring activity.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the instrument, except where magnetic materials are essential.

3.3.3 Nonferrous materials. Nonferrous materials shall be used for all parts of the instrument, except where ferrous materials are essential.

3.3.4 Metals. Metals shall be corrosion resistant or suitably treated to resist corrosion during storage or normal service.

3.3.4.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.5 Toxic and corrosive fumes. When subjected to the service conditions specified herein, materials and parts used in the construction of the instrument shall not liberate deleterious or corrosive fumes.

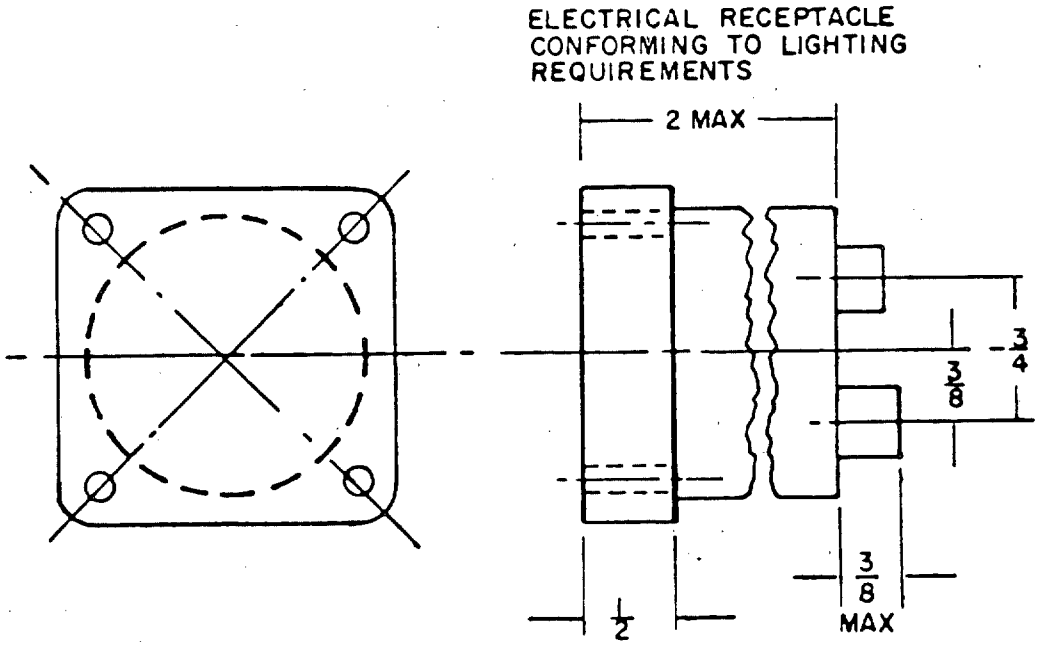
3.4 Design and construction. The altimeter shall be designed and constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Case. The case shall be in accordance with figure 1.

3.4.1.1 Body. The case body shall be constructed of nonferrous low density metal that is uniform in texture and shall have a smooth surface.

3.4.1.1.1 Static pressure connection. A static connection conforming to MS-33649-4 shall be placed in the rear of the case.

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CASE IN ACCORDANCE WITH MS 33638 (2-INCH) EXCEPT AS SHOWN

PRESSURE INSERT AND ELECTRICAL RECEPTACLE LOCATED ON VERTICAL CENTERLINE

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ± 1/32

FIGURE 1 CASE

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3.4.1.1.2 Reinforcement. The inside of the case around the pressure connection shall be of sufficient strength to prevent damage to the case when the connection is tightened during installation of the altimeter.

3.4.1.2 Bezel. The bezel shall be constructed of nonferrous low density metal. The bezel shall be held in place by screws properly secured by lock-washers or similar devices of approved design.

3.4.1.3 Cover glass. A cover glass conforming to MS 28105-11 shall be provided. It shall be properly sealed and replaceable by removing the bezel. If the cover glass is used as part of the lighting system, its size and shape may deviate from the dimensions specified in MS 28105 to the extent required for proper lighting design. If the cover glass and bezel form an integral assembly, they may be replaced as a unit.

3.4.1.4 Mounting lugs. Mounting lugs of the complete assembled case shall each withstand for 1 minute without fracture a load of 175 pounds applied along the mounting hole axis and toward the front of the instrument.

3.4.1.5 Shipping plug. A suitable shipping plug shall be screwed in the pressure insert for shipping purposes. The plug shall incorporate a vent consisting of a hole 1/16 ±1/64 inch in diameter.

3.4.2 Mechanism. The altimeter mechanism shall be a pressure actuated type. The design shall be reasonably simple to permit overhaul or repair without the use of special tools or fixtures.

3.4.2.1 The mechanism shall include a calibration adjustment to adjust or correct the indications to conform to the performance required herein. The calibration adjustment shall not affect the temperature performance of the instrument and shall be simply manipulated without the use of special tools.

3.4.3 Dial and pointer. The dial and pointer shall be fabricated of a material that will provide adequate illumination under night lighting conditions.

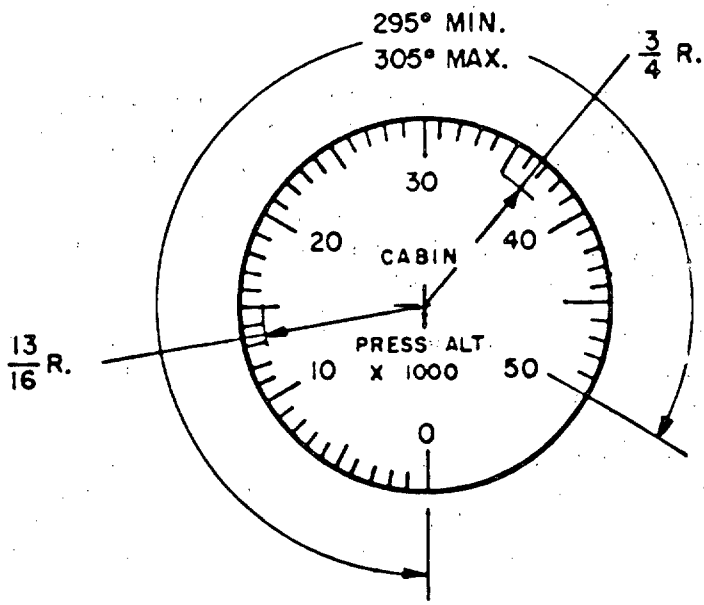
3.4.3.1 Dial. The dial shall be in accordance with figure 2.

3.4.3.2 Pointer. The pointer dimensions shall be in accordance with MS33585-1.

3.4.4 Distance from dial to cover glass. The distance from the dial to the cover glass shall be as small as practicable and shall not exceed 0.125 inch.

3.4.5 Integral lighting. The altimeter shall be integrally lighted with two parallel lighting circuits as shown on figure 3. The red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-L-27160. In the event of conflict between this specification and either MIL-L-25467 or MIL-L-27160, this specification shall govern. Although resistors shall not be permitted in the original design of the lighting systems, space shall be allocated for incorporation of a trimming resistor in each lighting circuit in the altimeter. The magnitude of the trimming resistors shall be specified by the procuring activity in terms of the brightness decrease required. The lighting shall be subject to the approval by the procuring activity.

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NOTE: ALL GRADUATIONS ARE PARTIAL RADIAL LINES.

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ± 1/64
ANGLES ± 0°15'

FIGURE 2. DIAL

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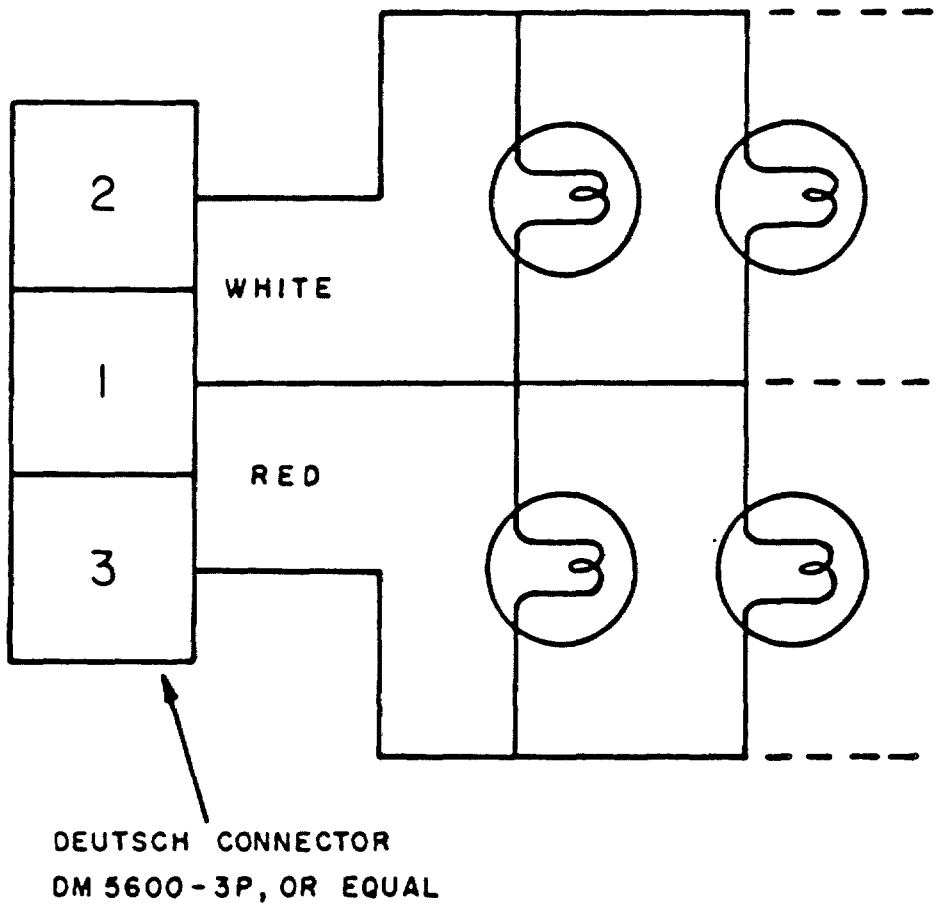


FIGURE 3. INTEGRAL LIGHTING CONNECTOR

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3.4.5.1 Prime standard altimeter. The integral lighting shall be comparable to and shall balance in brightness with a prime standard altimeter. The prime standard shall be an altimeter of the type specified herein. The prime standard shall have applied to its lighting circuits the specific voltage that the procuring activity has established as being necessary to assure equality of presentation with respect to the lighting with adjacent but dissimilar instruments. The contractor shall submit the test procedure for determining compliance with this requirement to the procuring activity for approval.

3.4.6 Screw threads. Screw threads shall conform to MIL-S-7742.

3.4.7 Mounting screws. Mounting screws for installing the altimeter shall be type 6-32NC-2A roundhead brass machine screws and shall be of sufficient length for mounting on a panel 3/8 inch thick. Screw finish shall be dull black.

3.4.7.1 A sufficient number of mounting screws shall be packaged with each altimeter and furnished in an envelope marked as follows:

" IMPORTANT
THIS ENVELOPE CONTAINS MOUNTING SCREWS "

3.5 Performance. The altimeter shall meet the performance specified herein when subjected to the following conditions:

a. Temperature: Temperatures ranging from -65°C to +71°C

b. Humidity: Relative humidity up to 100 percent.

c. Altitude: Altitudes ranging from 30 to 3.424 inches Hg

d. Vibration

(1) Vibration error: Vibration of 0.018 to 0.020 inch at frequencies from 500 to 3,000 cpm

(2) Vibration endurance: Vibration of 0.018 to 0.020 inch at a frequency of 2,000 cpm for a period of 3 hours

e. Case leakage: A vacuum sufficient to cause a change of reading of the pointer of approximately 18,000 feet.

f. Hysteresis: Hysteresis effects as will be encountered in service.

g. Scale error: Pressure over the scale range without exceeding the specified tolerances

h. Position error: Tapping of the indicator while operating in various positions with the pointer indication not varying more than 200 feet.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.7 Weight. The weight of the altimeter shall not exceed 0.6 pounds.

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3.8 Finishes and protective coatings

3.8.1 Aluminum alloy parts. Aluminum alloys parts shall be covered with an anodic film conforming to MIL-S-8625 except as follows:

3.8.1.1 Dials, small holes, and case inserts need not be anodized.

3.8.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.8.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatment in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.8.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.8.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.8.2 Steel parts. Steel parts shall be zinc plated where applicable. Zinc plating shall be in accordance with ASTM B-633-78 as applicable, and of a class that is adequate to achieve the degree of protection required.

3.8.3 The case body, the bezel, the unshaded portion of the pointer, and the background of the dial shall be finished in durable dull black, color number 37038 of FED-STD-595.

3.8.4 The markings specified in 3.9 shall be finished in white, color number 37875 of FED-STD-595.

3.8.5 Anti-reflection coating (AAV-3A/A). All reflecting surfaces shall be provided with a reflection-reducing coating that withstands the environmental conditions specified herein and meets the requirements of MIL-C-675, except that the following reflectance tolerances shall apply:

Angle of Incidence	Wavelength (Millimicrons)	Percent Reflectance
0°	450 to 675	0.6 Absolute
0°	425 to 700	0.5 Average
30°	450 to 625	1.0 Absolute
30°	425 to 700	0.5 Average

3.8.6 Protective treatment. When materials and parts are used in the construction of the instrument that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

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3.9 Markings. All of the following markings shall be durable to withstand usage encountered in service. The form of the letters and numerals on the dial shall conform to MS33558.

Marking	Height or Length ±0.010	Width of Line of Graduation ±0.005
Numerals 0, 10, 20, 30, 40, 50	0.156	------
Graduations corresponding to "0" and each subsequent 5,000 foot graduation	As shown on figure 2	0.020
All other graduations	As shown on figure 2	0.012
Lettering "CABIN PRESS ALT"	0.093	
Lettering "X 1000"	0.62	

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.11 Workmanship

3.11.1 General. The altimeter shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assembly, welding, brazing, and freedom of parts from burrs and sharp edges.

3.11.2 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with best shop practices. Where dimensions and tolerances affect the interchangeability, operation, or performance of the altimeter, they shall be held or limited accordingly.

3.11.3 Screw assemblies. Assembly screws and bolts shall be tight. The work "tight" means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt threads.

3.11.4 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.11.5 Gears. Gear assemblies shall be properly aligned and meshed, and shall operate without interference, tight spots, loose spots, or other irregularities. Where required, for accurate adjustment, gear assemblies shall be free from back lash.

3.11.6 Cleaning. Units shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, or other foreign material after assembly. Burrs and sharp edges as well as resin flash which might crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance

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of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the altimeter shall be classified as follows:

a. Qualification testing

b. Acceptance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Tapping. Unless otherwise specified, the altimeter shall be tested in normal operating position and shall be lightly tapped before a test reading is taken. (see 6.2.1.)

4.3.3 Vibration. A vibration stand shall be used that will vibrate at any desired frequency between 500 and 3,000 cycles per minute and will so subject the altimeter to vibration that a point on the instrument case will describe, in a plane inclined 45° to the horizontal plane, a circle of the diameter specified herein.

4.4 Qualification testing.

4.4.1 Test samples. The qualification test samples shall consist of two altimeters representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6, "Test methods."

4.4.3 Test report. After completion of the qualification tests, the contractor shall prepare a qualification test report in accordance with MIL-STD-831 and submit three copies to the procuring activity for approval.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling plan A.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6, "Test methods."

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a. Examination of product

b. Case leakage

c. Scale error - room temperature

d. Hysteresis

e. Individual tests of MIL-L-25467 and MIL-L-27160

4.5.2 Sampling plans and tests

4.5.1 Sampling plan A. One altimeter selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests as described under 4.6, "Test methods."

a. Individual tests

b. Scale error (-55°C)

c. Position error

d. High temperature characteristics

e. Vibration error

f. Friction

g. Overpressure

h. Sampling plan A tests of MIL-L-25467 and MIL-L-27160.

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 altimeters selected at random from the first 15 items of the contract or order shall be subjected to the following tests as described under 4.6 "Test methods":

a. Sampling plan A tests

b. Magnetic effect

c. Vibration endurance

d. High altitude - low temperature

e. Humidity

f. Mounting lugs

g. Sampling plan B tests of MIL-L-25467 and MIL-L-27160.

4.5.2.2.1 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent of cause and failure are determined.

4.5.2.2.1.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced

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shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The altimeter shall be inspected to determine compliance with this specification with respect to materials, workmanship, and marking.

4.6.2 Case leakage. A vacuum sufficient to produce a change of reading of the pointer of approximately 18,000 feet shall be slowly applied to the static pressure connection of the altimeter at which point the connection tubing shall be pinched off or otherwise completely sealed. The change in reading in 1 minute shall not exceed 500 feet.

4.6.3 Scale error-room temperature. The altimeter shall be subjected successively to the pressures specified in table I. The reduction in pressure, as indicated by a standard mercurial barometer during the interval between test points, shall be made at a rate corresponding to an increase in altitude of approximately 3,000 feet per minute. The altimeter shall remain at the pressure corresponding to each point for at least 1 minute but not more than 10 minutes before a test reading is taken. The scale error shall be tabulated. Movement of the pointer shall be free from backlash and any irregular action when the pressure is varied uniformly. The scale error at any point shall not exceed the tolerance listed in table I.

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TABLE I. Scale error test.

Standard Altitudes	Equivalent Atmospheric Pressure	Scale Error Tolerance (feet)	
		Room Temperature (Variation from Standard Altitude)	-55°C (Variation from Room Temperature Reading)
Feet X 1000	Inches of Mercury	Feet x 100	Feet X 100
0	29.92	2	3
5	24.89	3	3
10	20.58	4	4
15	16.88	5	5
20	13.75	6	6
25	11.10	7	7
30	8.88	8	8
35	7.04	9	9
40	5.54	10	10
45	4.36	11	11
50	3.42	12	12

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4.6.4 Hysteresis. Not more than 15 minutes after the altimeter has been first subjected to the pressure corresponding to the upper limit of the scale in the scale error test, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 3,000 feet per minute until the pressure corresponding to the first test point given in table II is reached. The altimeter shall remain at this pressure for at least 5 minutes, but not more than 15 minutes, before the test reading is taken. After the reading has been taken the pressure shall be further increased at the above rate until the pressure corresponding to the second test point given in table II is reached. The altimeter shall remain at this pressure for at least 1 minute but not more than 10 minutes before the test reading is taken. After the reading has been taken, the pressure shall be further increased at the above rate until atmospheric pressure is reached. The reading of the altimeter at either the two points shall not differ from the reading of the altimeter for the corresponding altitude in the scale error test by more than the tolerance given in table II.

TABLE II. Hysteresis.

Test Points	Standard Altitude (feet)	Equivalent Atmospheric Pressure (In. Hq)	Permissible Variation (feet)
First Point	25,000	11.10	300
Second Point	20,000	13.75	300
Third Point	---	Ground Level	150

4.6.4.1 After effect. Not less than 1 minute and not more than 5 minutes after the completion of the hysteresis test, the pointer shall have returned to its original reading, corrected for any change in atmospheric pressure, within the tolerance given in table II at the third point.

4.6.5 Scale error (-55°C). For a period of not less than 12 hours prior to this test, the altimeter shall not have been subjected to any test involving operation at other than atmospheric pressures. The test specified in 4.6.3 shall be repeated with the difference that the temperature of the altimeter during the test shall in this case be approximately -55°C, this temperature having been maintained for a period of 3 hours. The test points shall be those given in table I. The change in scale errors of this test from corresponding scale errors in the room temperature test shall not exceed the tolerance specified in table I.

4.6.6 Position error. At normal atmospheric pressure, readings shall be taken while the altimeter is being tapped lightly in each of several different positions. The change in pointer indication with change in altimeter position shall not exceed 200 feet.

4.6.7 High temperature characteristics. The altimeter shall be subjected to a temperature of +71°C for a period of 3 hours. It shall then be allowed to cool to a normal temperature, after which it shall be subjected to the tests specified in 4.6.3 and 4.6.2. The altimeter shall meet the requirements specified for these tests.

4.6.8 Vibration error. The altimeter shall be mounted on a vibration stand with a line joining the centers of the two upper mounting holes in a horizontal plane and with the plane of the mounting lugs of the altimeter

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vertical. The altimeter shall be subjected to vibration with circle diameter between 0.018 and 0.020 inch at frequencies from 500 to 3,000 cycles per minute to determine that the natural frequency of the altimeter does not occur in this frequency range. While the altimeter is being vibrated at frequencies from 500 to 2,500 cycles per minute, the range of pointer oscillation shall not exceed 250 feet and the pointer variation from its original position shall not exceed 250 feet. While the altimeter is being vibrated at frequencies from 2,500 to 3,000 cycles per minute the range of pointer oscillation shall not exceed 400 feet and the pointer variation from its original position shall not exceed 400 feet.

4.6.9 Friction. The altimeter shall be tested for friction at each of the test points listed in table III. The pressure shall be decreased to bring the pointer approximately to the desired reading, then held constant while two readings are taken; the first before the altimeter is tapped, the second, after tapping. The difference of the two readings shall be recorded as friction and shall not exceed the tolerances specified in table III. This test may be combined with the test specified in 4.6.3.

TABLE III. Friction.

Altitude Feet x 1000	Tolerance Feet x 100
5	2
10	2
15	2
20	3
25	3
30	3
35	4
40	5
50	7

4.6.10 Overpressure. The altimeter shall be subjected to an absolute pressure of 50 inches Hg for a period of 1 hour. For a period of not less than 6 hours following this overpressure, the altimeter shall not be subjected to any operation other than atmospheric pressure. The scale errors shall not vary from the previously noted scale errors by more than the tolerances listed in table IV.

TABLE IV. Variation from original scale errors after subject to 50 inches hg pressure.

Stand Alt. Feet x 1000	Variation Feet x 100
0	2
5	2
10	2
20	2

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TABLE IV. Variation from original scale errors after subject to 50 inches hg pressure (continued).

Stand Alt. Feet x 1000	Variation Feet x 1000
25	2
30	2
35	3
40	3
45	4
50	4

4.6.11 Magnetic effect. The altimeter shall be held in various positions with its nearest part 5 inches from and magnetically east or west of the center of a short bar magnet compass. The compass deflection shall not exceed 1°. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted.

4.6.12 Vibration endurance. The scale error shall be noted at the existing atmospheric pressure. The altimeter shall be subject to vibration with a cycle diameter of 0.018 - 0.020 inch for a 3 hour period at a frequency of 2000 cycles per minute. At the end of the vibration period the scale error at atmospheric pressure shall be again noted. The difference between the reading determined after vibration, after correction for any changes in atmospheric pressure, shall not exceed 100 feet.

4.6.13 High altitude - low temperature. The altimeter shall be placed in a chamber and maintained at a temperature of approximately -65°C and a pressure equivalent to approximately 50,000 feet for 4 hours. The altimeter shall then be maintained at a temperature of approximately -65°C (atmospheric pressure) for an additional period of 44 hours. The indicator shall then be allowed to stablize at room temperature and meet the room temperature scale error test requirements specified in 4.6.3.

4.6.14 Humidity. The altimeter shall be subjected to a humidity test in accordance with MIL-STD-810D Method 507.2 Procedure III except for a period of 5 cycles. Within 1 hour after completion of the last cycle, a scale error at room temperature test shall be conducted. The scale error shall not exceed the room temperature scale error tolerances specified in table I. Between 24 and 48 hours after completion of the last cycle, the mechanism of the altimeter shall be examined for corrosion. There shall be no evidence of corrosion of the internal or external parts of the altimeter which will affect the performance.

4.6.15 Mounting lugs. The altimeter case, with mechanism removed, shall be mounted face downward on the movable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug or mounting nut and attached to the pull strap. A load of 175 pounds shall be applied along the axis toward the front of the case for a period of 1 minute. There shall be no fracture.

4.6.16. Integral lighting. Each lighting circuit shall be properly connected individually and tested for conformance to 3.4.5.

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4.6.17 Reflectance (AAU-3A/A). The cover glass shall meet the requirements of 3.8.5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C or Commercial as specified (see 6.3).

5.5.1 Level A

5.1.1.1 Cleaning. Altimeters shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Altimeters shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation applications. Preservatives shall not be used.

5.1.1.4 Unless otherwise specified (See 6.3), each altimeter shall be individually unit packed in accordance with Method II of MIL-P-116.

5.1.2 Level C. The Level C preservation for altimeters shall conform to the MIL-STD-794 requirements for this level.

5.1.3 Commercial. The commercial/industrial preservation of altimeters shall be in accordance with the requirements of ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, C or commercial, as specified (see 6.3).

5.2.1 Level A. The packaged altimeters shall be packed in containers as specified in MIL-STD-795 for Level A protection.

5.2.2 Level B. The packaged altimeters shall be packed in containers as specified in MIL-STD-794 for Level B protection.

5.2.3 Level C. The packaged altimeters shall be packed in containers as specified in MIL-STD-794 for Level C protection.

5.2.4 Commercial: The packaged altimeters shall be packed in accordance with the requirements of ASTM D 3951 .

5.3 Marking

5.3.1 Levels A, B and C. In addition to any special or other identification marking required by the contract (See 6.3), each unit pack, intermediate and exterior container and utilized load shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial/industrial marking shall be in accordance with the requirements of ASTM D 3951 .

6. NOTES

6.1 Intended use. The altimeter covered by this specification is intended for use in aircraft equipped with pressurized cabins to indicate the pressure altitude in the cabins.

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6.2 Definition

6.2.1 Tapping. The phrase "Tapped Lightly" is defined as meaning the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.3 Ordering data.

6.3.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification

b. The level of packaging and packing desired

c. Whether sampling Plan B tests are to be conducted.

d. Type of altimeter required (see 1.2).

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is San Antonio ALC/MMIRAC, Kelly AFB, Texas 78241 and information pertaining to qualification of products may be obtained from that activity.

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Air Force - 99

Review : Activity

DLA-GS

Preparing activity:

Air Force - 82

(Project Number 6685-F733)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-25933B(USAF)	Altimeter, Pressurized Compartment integrally lighted, .0-50,000 feet, 1 7/8 Inch Dial

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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